

How Safety is Learned: Trajectories of Location Sharing Practices from Childhood into Young Adulthood

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Location sharing has increasingly become an integrated part of mobile phone use. In this paper we present a study that investigated the trajectories of how parental strategies of providing youth with smartphones in childhood led to long term impact on youth's own location-sharing behavior in adulthood. Through retrospective interviews with 17 emerging adults (18-22), we found that most of their parents took a gradual approach, sometimes starting with basic phones to help their children adapt. As youth transitioned to full smartphones, tensions arose—parents used them for easier communication and safety, while youth primarily saw them as social tools. Our findings revealed that youth tracked by parents during childhood often continued sharing their location in college, despite discomfort, and also shared with others for safety reasons. These results highlight the need for a balance between youth privacy and parental peace of mind in location-based technologies.

CCS Concepts: • **Human-centered computing** → **Empirical studies in collaborative and social computing**.

Additional Key Words and Phrases: Location sharing, Mobile phones, teens, family, safety, privacy

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1 Introduction

The prevalence of mobile phones, especially smartphones, has led to widespread debate around parental oversight and teen children's autonomy. While parents of teenagers have always had a desire to know *where* their children are, at any given time, mobile phones (in general) served as a game changing tool for information and control, first as a simple real-time communication tool, second, as an automatic tracking tool. Public media has increasingly questioned the right level of control, and research has sought to determine the balance between utility and limitations, for different groups of youth, ranging from young children, adolescents, and emerging adults [38]. However, much of the prior work has focused on the adolescent phase [4, 22, 23], leaving the

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long-term implications of parental mediation, particularly on digital location-sharing practices in adulthood, underexplored. The current college generation, emerging adults between 18 and 25, engenders a unique set of people, whose first experiences with mobile phones were with smartphones. This narrow slice of a now adult generation Z had access to location-sharing applications and social media from the get-go of their mobile phone ownership (in 2018, 95% of US based teenagers reportedly owned smartphones [6]). While mobile phone ownership among children and teenagers sharply increased in the two-thousand teens [20, 26], and parental controls and approaches have been well documented and researched, we have yet to understand how specific uses and practices during adolescence relate to the practices this generation now aligns with, as young adults.

Focusing on direct, continuous location sharing, we took a retrospective approach and asked now-college aged smartphone users who currently share location with others, to look back at how they initially were provided with a mobile phone, how they used it as teenagers living at home, and how they transitioned out of their home, to a college setting. We focused particularly on location-sharing with parents, a functionality that many of them leveraged with parents and friends from the early days of their mobile phone ownership. This provided a groundwork analysis for us to start understanding how and why location tracking (or sharing) was initiated (or not) between the youths and their parents. We then explored how tensions between their parents and themselves developed over the years, how these tensions may have shifted, if at all, between when they lived with their parents, and as they moved to dormitories or shared apartments in college. This focus on location-sharing during the transition to adulthood emphasizes an underexplored angle in parental mediation research, shedding light on its long-term effects. Through this research, we aimed to answer the following research questions:

- RQ1:** *What motivations and arguments were youths given by their parents, when they were first given smartphones?*
- RQ2:** *How did early parental mediation influence youth's location sharing practices during their transition from childhood to adulthood?*

Overall, most participants were introduced to mobile phones gradually, often starting with basic phones or smartphones without cell service, useful only with known wifi. Parents prioritized ease of communication and physical safety, while youth primarily used phones for social connection (RQ1). Location-sharing behaviors varied based on parental mediation during childhood. Youth whose locations were tracked by parents were more likely to continue sharing their location with parents in college, while those whose parents relied on manual check-ins were less likely to do so. Youth who continued sharing their locations with parents in college also shared with other close relations such as roommates and significant others for safety purposes (RQ2).

Our research makes a novel contribution to the broader HCI community by shedding light on the developmental trajectory of location-sharing practices, an underexplored facet of parental mediation in digital technology use. By examining how the reported parental strategies during adolescence influence emerging adults' location-sharing behaviors, our work provides critical insights into the evolving interplay between social and technical factors. The qualitative analysis, rooted in retrospective interviews, captures the personal experiences of participants, offering a better understanding of how family dynamics and trust mediate the adoption and adaptation of digital tools like location tracking apps and smartphones themselves. These findings not only deepen our understanding of family dynamics when it comes to parental monitoring but also inform the design of adaptive mobile technologies that address the shifting needs of individuals as they transition through life stages.

2 Related Literature

A given premise for real-time location awareness by parents of their independent teenage children is a mobile phone. Before mobile phones became readily available, people's whereabouts were mostly communicated in advance or through landlines. This setup did not provide for much flexibility in terms of updating whereabouts (e.g. for coordinating pickup, meals, reporting late arrival, or simply letting the other part know where they are); people above the age of 25 at the time of this writing have lived in such a world, where having real-time knowledge of exact location of loved ones was not always an option (obviously, people often did know where each other were, but updating depended on the availability of landlines). Mobile phones and later smartphones changed all of that. And while mobile phones were initially envisioned as a business tool for a minority adult group of professionals, the early 2000s saw a spike in adoption of mobile phones, pushed by lower service prices and subsidized handsets.

In particular, smartphones emerging in the late 2000s were adopted by teenagers at a faster pace than by adults, with social media playing a major role [20]. By 2018 (when our participants would have been between 13 and 17 years old), 95% of U.S. teenagers owned a smartphone [8]. Research in relation to teenagers and smartphones consequently took off and many studies have focused on the emerging harm that the always-on, always-close devices potentially have on mental health [21], academic performance [24], identity building [30] and their social connections [1]. Additionally, smartphone use has sparked conflict between parents and teens, with one study for example revealing that parents' lack of control over their own smartphone use, contributed to parent-child conflict [29]. Since location-tracking is often presented as a safety tool, particularly towards teenagers, the notion of digital safety emerged as social media enabled, particularly for underage children, online communication and interaction with strangers. With smartphones the concerns became exacerbated and a significant number of studies on online safety exists [5, 7, 14, 16, 28, 31, 34], particularly for children and teenagers. These studies addressed a multitude of different types of threats, from unwanted online sexual advances, cyberbullying and blackmailing, to stalking and online assaults.

Sharing continuous location digitally, is one way by parents to monitor and enable control; while early studies looked mostly at adults sharing with each other [9, 11, 35], more recent research has looked at location-sharing between parent and teen dyads. A review of 23 papers about location tracking described location-tracking as a practice that became common among parents [18], with motives ranging from safety, control, worry and a sense of security, to convenience, care, connection and curiosity. Reasons to refrain from location-tracking were trust, self-direction, privacy, and a lack of usefulness [18]. The review focuses solely on a *one-way tracking* perspective, not addressing how most relevant apps are two-way systems where parents' location is also available to the child.

Studies in Northern European countries on pre-teens indicate that parents monitor their child's location for assurance [37], to guide them [32], and for convenience in the daily organization of family life [19]. In terms of the US, other recent studies provide insights into location-tracking of teenagers. About half of a large US sample of adolescent-parent dyads reports the use of location tracking apps [13]. Similar to the research on pre-teens, a US cross-sectional survey study from 2004 among students also reports that digital location tracking is a two-way process. Parents of 47% of the college students tracked their child's location (at the time of study or in the past). Many of these students (a third of the total sample) also track(ed) their parents' location. The main reason for location tracking was safety. While the findings suggest that college students are more likely to perceive their parents as inhibiting their autonomy and being intrusive when they are tracking their locations, these effect sizes are modest [15].

While several studies have addressed location-sharing between parents and teenagers, as well as college students, no research has looked at the changes of such dynamic from the context of living at home to being independently residing elsewhere. This transitional perspective has been largely overlooked despite its significance in understanding evolving location-sharing practices. Our study seeks to fill this gap by tracing the reported practices from when the young adults were pre-teens and teenagers, to their current location-sharing practices, including with whom and for what reasons.

3 Methods

3.1 A Retrospective Interview Study with Emerging Adults

We conducted a retrospective, semi-structured interview study with 17 emerging adults, between the ages of 18 and 22 years old, residing in the US. These students represent a generation where smartphones have been available to them from the start, as opposed to earlier generations who had phones with only limited capabilities (such as calling and texting) as their first phones. We developed an interview protocol based on the qualitative research recommendations provided by Myers [33] as well as insight from a pilot session, allowing us to refine interview questions for clarity and relevance. Each interview session began with an exploration of the participant's background, including their upbringing, living arrangements, and their parents' parenting styles. We then delved into their experiences of adopting and using their first smartphone. To contextualize shifts in location-sharing behaviors, we also discussed their experiences with parental coordination and location-sharing habits during their middle- and high school years, as well as how these practices evolved after transitioning to university life. This study was part of a research project focusing on locations-sharing practices and the data presented in this paper represent a subset of the larger dataset. A related study, based on a larger dataset, focuses on how college students use location-sharing to create a sense of safety with their current local friends [10].

3.2 Participant Recruitment and Data Analysis Approach

We recruited emerging adults who were sharing location currently (either through the app Life 360 or Apple's built-in app Find My), with either friends, partner or parents, through word-of-mouth at several different universities: Personal connections were utilized, such as asking acquaintances to contact people with connections to universities, either by being students themselves or working at a university in some capacity. None of the authors had any direct connections to the participants, neither as teachers, nor being friends with their teachers or advisors. All participants were located in an urban area, for location-sharing to be more relevant to them. An urban setup leads students off-campus to a much higher degree than for example students in college-towns. The interview sessions (N=17) were conducted remotely via video conferencing and were both audio- and video-recorded. The study was approved by the second author's main affiliation through that university's standard ethics process and an alliance agreement was created between the other authors' affiliations at the time of the study. Participants were compensated \$25 for their time. The participants were between 18 and 22 and all lived in an urban area in the Northeastern United States. They had moved to the area for university and all but two came from different states. N = 11 youth self-reported as females, and N = 6 identified as males. They represented five different universities and were all undergraduate students. All participants lived away from their parents, either in dormitories or private/shared apartments. Each interview lasted approximately one hour.

After completing data collection, we conducted a thematic analysis following the guidelines outlined by Braun and Clarke [12]. This approach allowed for the identification of emergent themes while providing flexibility for new patterns to surface. The process began with transcribing the

audio recordings and familiarizing ourselves with the data by thoroughly reading the transcripts. Three of the authors collaborated to create an initial set of codes based on recurring topics within the transcripts. Using these initial codes, the first author systematically coded each transcript, allowing new codes to emerge throughout the process. When a new code was identified, all previously coded transcripts were revisited to ensure consistency and comprehensive application. Once coding was complete, the first three authors worked collaboratively to organize the codes into cohesive themes that aligned with our study's overarching research questions.

4 Results

In this section, we present the themes that emerged from our analysis of youth experiences with smartphone access and location-sharing during both childhood and adulthood. Quotations from participants are identified by their IDs (P1, P2, ..., P17), along with their age and gender. To provide a foundation for our results focusing on location-sharing, and to answer RQ1, we asked into our participants' initial acquisition of smartphones; pre-teens and teenagers depend on parents or caretakers providing them with a phone and as such, the digital communication dynamics between parents and their children started here. We realize that only talking to teens/young adults does not provide a full picture of motivations for giving the teens a phone; instead we gain an understanding of how the teens themselves understood their parents' motivations.

4.1 Factors Influencing Parental Decisions on Smartphone Provision During Childhood

While most participants received smartphones at a young age, their parents introduced them to this technology gradually. Nearly all participants (N = 16) reported receiving a smartphone in middle school (age eleven through thirteen), with only one participant receiving theirs in high school. However, the majority of participants (N = 10) noted that before receiving a smartphone, they were first given a non-smartphone device. This gradual approach often involved providing a phone without cell service, limiting its use to Wi-Fi networks and preventing regular phone calls. Despite this restriction, these phones were still capable of accessing the internet, playing games, and sending messages when connected to Wi-Fi. Participants also shared that when older siblings upgraded to new phones, parents would often pass down their old, SIM card-less (i.e. without cell service) devices to the youth. In contrast, seven participants were given flip phones or basic button phones before eventually transitioning to smartphones.

"I had another cell phone in 4th grade where it was like one of those really old phones, but like the keypad ones. Where I had to type multiple times to get a letter." – P6, Female, 19 years old

The remaining participants (N = 7) reported that their first phone was a fully-functional smartphone. They often reflected on the online risks they faced due to having unrestricted internet access at a young age. However, they also acknowledged that this early exposure taught them valuable lessons about setting boundaries in social interactions, helping them navigate digital spaces more responsibly than they might have otherwise.

"I got a smartphone when I was in 6th grade. I thought I was gonna get a flip phone because that's what they [older siblings] had had growing up. So I was asking my dad for it for Christmas and he just gave me an iPhone. ... I don't really think at that age I should've really had unlimited access honestly. It's not that it did anything bad for me, obviously, I feel like it was a lot for an 11 year old and at that time nobody really knew how much stuff I was kind of getting into. But [...] I learned very quickly the boundaries like, how to treat people on social media." – P14, Female, 20 years old

When giving smartphones to children, parents primarily aimed to ensure easy communication and safety, while youth mainly used them for socializing. Most participants (N = 11) reported that their parents provided them with smartphones to maintain **regular contact**. As these youth entered middle school and became more independent with after-school activities or visits to friends' houses, parents saw smartphones as a practical tool to stay connected and ensure their children could reach out when needed.

"It was just for being able to keep in touch with my parents, 'cause I remember I was one of the first people to get a phone at my age. Like my friends didn't even have one. So, it was just for like my parents, a way for me to contact them ... whenever I would go out." – P6, Female, 19 years old

More than half of the participants (N = 10) also noted that their parents gave them smartphones to **keep track of their whereabouts**. Many of the youth shared that their parents would frequently call or text them from work to check where they were, when they would leave for after-school activities, or whether they had returned home safely. Some of these youth mentioned feeling obligated to update their parents on their location every time they moved, as described by P3 below:

"To tell them where I was. So, from middle school and high school, every time I got home, I would text 'em and just say, 'Hey, I'm home,' or, 'I'm going to friend's house, so if you get home before me, you know where I am.' So yeah, pretty much when I went anywhere that was kind of even just across the street, I'd let them know." – P3, Male, 22 years old

Some youth (N = 4) also mentioned that their parents gave them smartphones primarily to help **coordinate pickup and drop-off logistics**. These youth relied on their parents for transportation to and from after-school activities, and having a phone facilitated smoother communication for planning rides. Lastly, a few participants (N = 4) expressed that they felt compelled to get a smartphone because all of their friends had one. The desire to fit in socially led them to request a smartphone from their parents.

"It was like everyone around me had one. So I wanted to be included in that." – P16, Female, 20 years old

While parents carefully considered various factors when deciding to give their children smartphones, youth primarily used these devices for social connections. Two-thirds of the participants (N = 12) reported that their primary use of smartphones was to stay connected with friends through online communication. Most participants relied on texting or instant messaging as their preferred method of keeping in touch, though a few also mentioned using platforms like FaceTime.

"It was just texting, kind of whoever had a phone at the time. Sometimes, if I was lazy enough to not walk over to my friend's house,... I could just text like, 'Hey, what are you doing?'" It was mostly texting, and then facetime if needed." – P3, Female, 22 years old

Approximately half of the participants (N = 8) reported using their smartphones primarily to access social media. Only two of these mentioned seeking parental permission before installing social media apps, and one had restricted online access. Interestingly, peer influence was frequently cited as a major factor in their decision enter the social media space.

"Social media started taking off in my town around that time. So I was able to get in contact with people I wasn't super close with also, getting into Instagram and stuff. So that was the main thing that I used." – P8, Female, 20 years old

Some other participants (N = 8) reported using their smartphones primarily for entertainment, such as playing games and listening to music. Some specifically mentioned using their smartphones to explore personal interests and hobbies, with photography being a notable example. The convenience of taking photos with a smartphone camera, combined with the ability to instantly share

them, made this hobby more accessible. Platforms like Instagram and Snapchat were frequently cited as key outlets for sharing images, enabling them to express their creativity and receive feedback from peers.

“I think definitely to take photos of everything, everywhere. But, I was on Instagram like as everyone... I’m in such a weird part of the generation that’s like, we’re not too much of iPad babies like crazy and on top of attached to phones and stuff. Then I would just Instagram.” – P3, Female, 22 years old

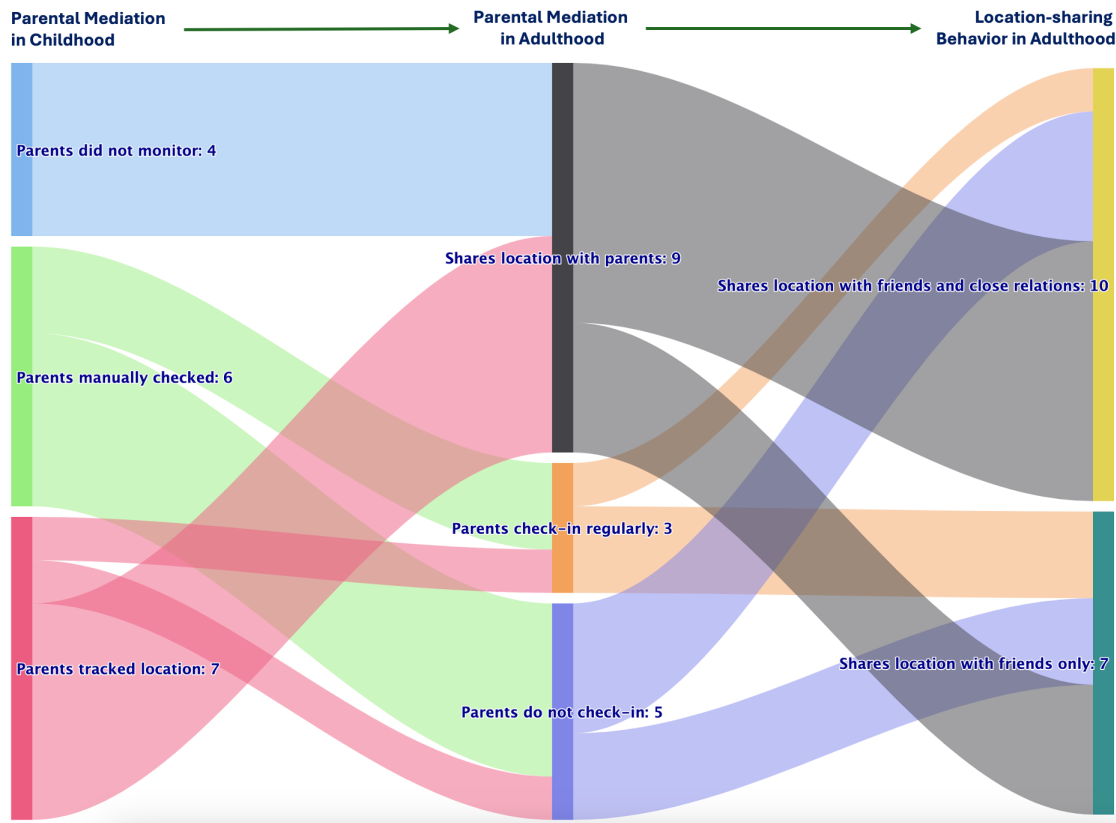


Fig. 1. Sankey Visualization of Parental Mediation and Youth’s location-sharing Behavior

4.2 The Influence of Parental Mediation on Youth Location-Sharing Practices During Adulthood

As the youths grew older and had to navigate new social contexts, the parental strategies and mediation practices they encountered during childhood continued to influence their decisions about location-sharing and safety. This section focuses on how these early parental mediation practices impacted youths’ location-sharing behaviors in adulthood, and explores the nuanced interplay between past experiences and present decisions. To aid this analysis and description, we developed a Sankey diagram (Figure 1) that visualizes the flow of relationships between structured codes, mapping the value transitions as outlined by Schmidt [36]. The three vertical bars represent youth classifications based on parental mediation when they were children (left), parental mediation when

they became adults (middle), and their location-sharing behavior with other relations currently (right). Reading from left to right, the diagram highlights the connections between youths' parental mediation during childhood and its continuation into adulthood. Conversely, reading from right to left, it illustrates the relationship between youths' location-sharing behavior with their friends and other close relations, such as significant others and roommates, and the degree of parental mediation in adulthood. Our findings suggest that youths' location-sharing behavior with their parents and other close relations in adulthood was influenced by the parental mediation they had during childhood. Below, we examine these dynamics in detail.

Youth who were tracked by their parents during their childhood were more likely to continue sharing their location with parents in adulthood. Approximately half of the participants ($N = 7$) reported that their parents had tracked their locations during childhood using apps like Find My or Life 360. Of these, most participants continued sharing their locations with their parents in college as they moved to new cities. Our participants emphasized how their parents had had open discussions with them about sharing and the participants were aware that the continuation was primarily for their own safety. However, this form of tracking sometimes led to conflicts as some participants would occasionally lie about their whereabouts, prompting confrontational calls or texts from their parents. Others described creative workarounds, such as linking location-sharing services to stationary devices like laptops or tablets left at home, enabling them to bypass real-time tracking. Despite these, all participants acknowledged that their parents' intent was to ensure their safety.

[My cousin] once had brought me into San Francisco to visit college friends. And he had gone to meet up with coworkers. And it was like getting late and later, and it started approaching like midnight and then 1:00. Um, and it [location-sharing app] had kind of relayed the information to my mom, and then every hour, she'd be like, "Where are you now? Why are you not in [college city] yet? Why are you there?" Like, "What's going on?" And I was like, "Don't worry," like, "All is safe." Which, I get it, like me being in a new place, nerves and everything, but I was like, at that point, even if I do let you know exactly where I am, it's not gonna make a difference. - P13, Male, 20 years old

Another theme we observed among youth whose parents tracked their locations during childhood was how they extended their location-sharing practices beyond their parents, as they moved to college. Many began sharing their locations with close connections such as roommates, romantic partners, or close friends. While this was often because of social purposes — keeping up with each other's outings or travel plans - nearly all of them emphasized that safety was a key motivator. Notably, the youth themselves often initiated this sharing practice, recognizing the need for added safety measures.

"And then coming to college, my roommate and I made really good friends, and I started sharing our locations. And that was mainly for safety reasons, just because we're in [big city]. And then if anyone's out walking late at night by themselves or something, you could just send a text to the group chat like, hey, just a heads up, I'm by myself. If you could just check my location occasionally, check if I get home and something like that. That definitely became a bigger thing in college." - P15, Female, 20 years old

Youth whose parents relied on phone calls or texts to manually check their whereabouts during childhood were less likely to share their location with parents in adulthood. A third of the participants ($N = 6$) reported that their parents, rather than using location-sharing apps, preferred calling or texting to monitor them. Most of them explained that this was because their parents were unfamiliar with these technologies. Ironically, several participants expressed that they would have preferred their parents to use location-sharing apps, as it would have been less intrusive than

the constant calls or texts about their whereabouts. Most of these participants also noted that their parents stopped checking in once they entered college. Youth also sometimes brought up that even if parents asked for location-sharing access in college, youth would refuse or limit that access.

“I think if they did [ask to share location], I would give it to them, but I would also then be more conscious of like, oh, if I’m going somewhere really late at night and I don’t want them to know about it, I would turn my location services off for them so like they wouldn’t receive the notifications.” – P6, Female, 19 years old

Youth who had a trust-based relationship with their parents, without location tracking or manual checking, during childhood, were more likely to share their location with parents during adulthood. This category of youths grew up in trust-based relationships with their parents, who were more lenient about their activities and whereabouts. Rather than tracking their location or calling frequently to check on them, parents trusted their children’s judgment when it came to going out and staying safe. However, once these youths entered college and moved to different cities, all of them began sharing their locations with their parents. Concerned for their safety, parents requested access to their location, and in many cases, the youth themselves initiated the safety measure, asking their parents to install location-sharing apps. Similar to the youth whose parents tracked their location in childhood, these participants also shared their location with their friends and close relations, such as significant others and roommates, as an additional safety measure.

“When I moved to the city I completely gave my mom permission to see my location and my dad. So they both currently have my location, ’cause I want them to have it. It’s a safety thing, I definitely live by myself... and so I share it with my roommate and then also my boyfriend.” – P9, Female, 18 years old

Overall, our findings revealed that youth who were tracked by their parents during their time in middle- or high school (via apps like Find My or Life 360) were more likely to continue sharing their location with their parents in college. While many youths acknowledged their parents’ concerns for their safety, location-sharing sometimes led to conflicts, with participants finding ways to bypass tracking when desired. Conversely, youths whose parents relied on manual methods—like phone calls or texts—were less likely to share their location in adulthood, yet still viewing location-sharing apps as less intrusive. Interestingly, youths who grew up with trust-based relationships, where parents did not track or check on their whereabouts, were more inclined to share their location with both parents and close relations once they moved to college, often initiating the use of location-sharing apps for safety and peace of mind.

5 Discussion

Our study highlights a developmental trajectory where early parental mediation methods—ranging from gradual smartphone introduction to explicit location tracking—continue to influence young adults’ safety practices. This documented trajectory aligns with prior research in the domain of adolescent online safety [2, 22, 38], which emphasizes the importance of parental mediation to ensure children’s safety online. However, our findings extend this body of work by demonstrating how early monitoring fosters a “safety culture” that persists into adulthood, albeit sometimes reluctantly. As such, early parental mediation shapes digital safety practices that persist and evolve beyond the parent-child relationship, extending into broader social contexts in adulthood. For instance, our participants frequently shared their locations with significant others and roommates during their adulthood, adapting parental oversight practices to fit peer networks. In our discussion we unpack this developmental trajectory into three themes, leading to broad suggestions for future research and design of location-sharing technology

5.1 Safety and Family Centric Technology

Although youth were not always comfortable with being tracked, they recognized the perceived safety it offered. Many of our participants noted that during their childhood their parents had open, transparent discussions about location-sharing, emphasizing its role in ensuring their safety. In many instances location-sharing had served as an invisible safety line that they did not need to pay a lot of active attention to. Prior research in the digital safety space [3, 22, 23, 38, 40] has shown that open communication between parents and children is crucial in fostering children's active engagement in their own online safety practices. Our results further reinforce this, showing that when parents address safety and privacy concerns collaboratively even during their childhood or adolescence, youth are more likely to understand, accept, and participate in these measures [23] even in adulthood. As such, we recommend that future researchers explore the design of family-centric technologies that include collaborative decision-making tools, enabling parents and youth to negotiate and openly discuss safety settings in real time, to emphasize a more balanced and participatory approach to safety management. Current systems often assume static relationships and use cases, but our findings highlight the need for designs that adapt to users' evolving roles and relationships. Future researchers should explore adaptive privacy settings, allowing users to customize sharing parameters based on the context, such as transitioning from familial to peer-based safety networks. Such tool would address our finding that some participants thought the parental location-sharing was invasive at their current adult stage in life, but could not find a more coarse grained way to continue sharing, and instead either left the fine-grained location-sharing on, or turned it off all-together.

5.2 Adaptive Location-Sharing Mechanisms

Additionally, our study found that parents introduced smartphones gradually during their childhood, often with the explicit goal of keeping track of them as they grew more independent. This aligns with the "caring dataveillance", a term proposed by Sukk and Siibak in 2021 [37], which described a way in which parents ensure safety and build digital literacy in their children. While these strategies might have provided parents with peace of mind, children sometimes also perceived such technological monitoring as an invasion of privacy, similarly to other research [3, 4, 22, 38, 39], which explains why some of our participants reported lying or finding workarounds to evade parental tracking. Such behaviors, in turn, can increase parental anxiety and potentially lead to unintended negative consequences for children. This aligns with prior research by Czeskis et al. [17] who examined parent-teen value tensions surrounding mobile safety technologies, emphasizing values of privacy, safety, trust and maturation. The study, which was conducted in 2010 before continuous location-sharing was widely available (and arguably accepted), probed into different kinds of information and different types of situations, asking dyads of teens and parents about sharing preferences. They found that both parents and teens preferred cooperative monitoring models where teens are notified and retain some control over when and how their information is shared, particularly distinguishing between emergency and non-emergency contexts. Our findings echo the study's emphasis on trust-based mediation but also reveal how these early patterns of parental mediation influence longitudinal safety practices into emerging adulthood, demonstrating that many participants continue voluntary location-sharing habits even after gaining independence in college. However, while Czeskis et al. concentrated on parent-teen dyads, our findings highlight how these early practices migrate into youth's peer networks, suggesting that safety norms established in adolescence reshape expectations for trust, transparency, and location visibility among friends, roommates, and romantic partners later in life.

As such, our study informs the design for adaptive location-sharing mechanisms that balance autonomy and safety, echoing the location-sharing framework for families proposed by Kayal et al. [25]. By incorporating hybrid models that combine manual check-ins and automated tracking, such tools can support context-sensitive usage. For example, mechanisms that let individuals choose when to share their location automatically and when to provide updates manually, depending on their preferences or the situation, may accommodate changing needs as individuals transition between childhood and adulthood or between families and peer networks.

5.3 Trust-Based Sharing

One of our key findings was that youth generally did not appreciate frequent parental check-ins via calls or texts during childhood, which often led to confrontations. By adulthood, youth resisted such invasive interactions. This highlights the importance of parents adopting less intrusive methods to ensure their children's safety while respecting their independence. The participants did in fact find the automated location-sharing they had used as teens to be less intrusive, well-knowing that their whereabouts would be of constant observation to their parents, if they checked. Yet, as long as it was built on trust, expressed as fewer explicit calls and texts to the theme of "where are you now?", they were less frustrated. Relatedly, research on adolescent online safety has shown that teens prefer safety mechanisms that offer autonomy and personal privacy [15, 35]. As such, trust-based interaction is important and incorporating non-intrusive reminders or prompts could facilitate trust-based interactions [17, 25]. Therefore, we recommend that future research focus on designing parental mediation technologies that align with these preferences, incorporating customizable features that balance parental oversight with youth privacy and agency, fostering both safety and independence. Location-sharing should be based on trust and not the other way around: building of trust with teen children through location-tracking, as some parents had practiced.

5.4 Limitations and Future Research

We conducted an in-depth retrospective interview study with a relatively small sample of emerging adults in the U.S. who currently share their location with others via their smartphones. A strength of this approach was our ability to deeply engage with their experiences of location-sharing services from the inception of becoming smartphone users to their transition into adulthood. Our study however presented several limitations, some related to generalizability: While we aimed to find a wide set of participants to include, our sample of college students all had a higher socioeconomic status and our findings may not be generalizable to all emerging adults, particularly those who come from different geographic locations, cultural contexts, socioeconomic backgrounds, or education levels, as these factors can have a strong influence on individual's technology adoption and usage behaviors. Our participants were all college students, a demographic with a likely higher socioeconomic status (SES) and education level than non-college bound emerging adults. These factors may have influenced their location-sharing technology usage, which were not explicitly addressed in our study design. Future research should include a more diverse participant pool, stratified by SES, education, and geography, for broader representation [27].

Further, excluding emerging adults who did not currently share their location with others might have constrained our findings. Additionally, the reliance on a small convenience sample limits the ability to draw broad conclusions, underscoring the need for larger-scale studies to better capture diverse behaviors and perspectives.

Therefore, future studies could for example take a longitudinal approach to tracking cohorts of youth across different life stages to understand these transitions at a larger scale, as well as study the contextual factors that may contribute to some adults choosing to not share their location with others, as studying non-use is as important as use. We also encourage future researchers to

explore the design of location-sharing tools that adapt to life transitions, offering features that accommodate changing privacy preferences. These designs should address the specific needs of diverse populations and consider disparities in SES and education levels. Additionally, research on non-users and partial adopters of location-sharing technologies would provide valuable insights to promote inclusivity and equity in design.

6 Conclusion

In this study we explored the use and past use of continuous location-sharing among young adult college students and their sharing trajectory from adolescence; we took a retrospective approach and asked now-college aged adults to look back at how they initially were provided with a mobile phone, how they used it as teenagers living at home, and how they transitioned out of their home, to live independently.

Revisiting our research questions, we start by addressing RQ1: *What motivations and arguments were youths given by their parents, when they were first given smartphones?* Majorly, youths were given smartphones to facilitate coordination with their parents, for entertainment and for the broadly defined notion of safety. This is in accordance with previous research and we simply reiterate this reasoning that serves as a foundation for our RQ2: *How did early parental mediation influence youth's location sharing practices during their transition from childhood to adulthood?* To this end, we documented how early parental mediation affected uses in adulthood: Youth who had their locations tracked by parents were more likely to continue sharing their location with parents in college, while those whose parents relied on manual check-ins were less likely to continue sharing. Youth who continued sharing their locations with parents in college also shared with other close relations such as roommates and significant others, motivated by safety purposes. As such, our research makes a novel contribution to the broader HCI community by shedding light on the developmental trajectory of location-sharing practices, an underexplored facet of parental mediation in digital technology use.

More broadly, our study illustrates the delicate balance between safety and privacy in parental oversight of smartphone use and location sharing from childhood to adulthood. As youth transition to more independent stages of life, we observed the benefits of parents' early involvement and guidance in their smartphone adoption. Early parental involvement in location sharing influenced youth's habits, with those whose parents monitored their location in childhood continuing to share their location with significant others or roommates in adulthood. However, there remains a need for broader adoption of such technology by parents, in which youth perceiving it as less intrusive. We encourage designers to prioritize non-invasive safety solutions that encourage trust and autonomy while maintaining security. By empowering both parents and youth through thoughtful location sharing technology, families can better navigate the complexities of understanding and negotiating safety in today's world.

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References

- [1] Michael Adorjan and Rosemary Ricciardelli. 2021. Smartphone and social media addiction: Exploring the perceptions and experiences of Canadian teenagers. *Canadian Review of Sociology/Revue canadienne de sociologie* 58, 1 (2021), 45–64.
- [2] Mamtaj Akter, Leena Alghamdi, Jess Kropczynski, Heather Richter Lipford, and Pamela J. Wisniewski. 2023. It Takes a Village: A Case for Including Extended Family Members in the Joint Oversight of Family-Based Privacy and Security for Mobile Smartphones. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI EA '23). Association for Computing Machinery, New York, NY, USA, Article 194, 7 pages. <https://doi.org/10.1145/3544549.3585904>
- [3] Mamtaj Akter, Leena Alghamdi, Jess Kropczynski, Heather Richter Lipford, and Pamela J. Wisniewski. 2023. It Takes a Village: A Case for Including Extended Family Members in the Joint Oversight of Family-based Privacy and Security for Mobile Smartphones. In *Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI EA '23). Association for Computing Machinery, New York, NY, USA, Article 194, 7 pages. <https://doi.org/10.1145/3544549.3585904>
- [4] Mamtaj Akter, Amy J. Godfrey, Jess Kropczynski, Heather R. Lipford, and Pamela J. Wisniewski. 2022. From Parental Control to Joint Family Oversight: Can Parents and Teens Manage Mobile Online Safety and Privacy as Equals? *Proc. ACM Hum.-Comput. Interact.* 6, CSCW1, Article 57 (apr 2022), 28 pages. <https://doi.org/10.1145/3512904>
- [5] Mamtaj Akter, Jinkyung Katie Park, Campbell Robinson Headrick, Xinru Page, and Pamela J Wisniewski. 2025. Calculating connection vs. risk: Understanding how youth negotiate digital privacy and security with peers online. *Proceedings of the ACM on Human-Computer Interaction* 9, 7 (2025), 1–26.
- [6] Chris Anderson. 2008. The end of theory: The data deluge makes the scientific method obsolete. *Wired magazine* 16, 7 (2008), 16–07.
- [7] Monica Anderson. 2016. Parents, Teens and Digital Monitoring. <https://www.pewresearch.org/internet/2016/01/07/parents-teens-and-digital-monitoring/>
- [8] Monika Anderson and Jingjing Jiang. 2018. Teens, Social Media and Technology 2018. (2018). <https://www.pewresearch.org/internet/2018/05/31/teens-social-media-technology-2018/>
- [9] Louise Barkhuus, Marek Bell, Scott Sherwood, Malcolm Hall, and Matthew Chalmers. 2008. From awareness to repartee: sharing location within social groups. In *proceedings of the SIGCHI conference on human factors in computing systems*. 497–506.
- [10] Louise Barkhuus, John S Seberger, Jorge P Campos, and Anouk Mols. 2026. Like Guardian Angels: Continuous Location-Sharing and the Production of Safety Among US College Students. *Computer Supported Cooperative Work (CSCW)* 35, 2 (2026), 7.
- [11] Julie Boesen, Jennifer A Rode, and Clara Mancini. 2010. The domestic panopticon: location tracking in families. In *Proceedings of the 12th ACM international conference on Ubiquitous computing*. 65–74.
- [12] Virginia Braun and Victoria Clarke. 2012. *Thematic analysis*. American Psychological Association.
- [13] Kaitlyn Burnell, Fernanda C Andrade, Sarah M Kwiatek, and Rick H Hoyle. 2023. Digital location tracking: A preliminary investigation of parents' use of digital technology to monitor their adolescent's location. *Journal of family psychology* 37, 4 (2023), 561.
- [14] Paolo Calciati, Alessandra Gorla, Konstantin Kuznetsov, and Andreas Zeller. [n. d.]. Automatically Granted Permissions in Android apps. <https://doi.org/10.1145/3379597.3387469>
- [15] Gregory E Chase, Morgan T Brown, Jessica L Navarro, Melissa A Lippold, and Michaeline Jensen. 2024. Digital location tracking in the parent/caregiver–college student dyad. *Journal of Adolescence* (2024).
- [16] Lorrie Faith Cranor, Adam L. Durity, Abigail Marsh, and Blase Ur. 2014. Parents' and Teens' Perspectives on Privacy In a Technology-Filled World. 19–35. <https://www.usenix.org/conference/soups2014/proceedings/presentation/cranor>
- [17] Alexei Czeskis, Ivayla Dermendjieva, Hussein Yapit, Alan Borning, Batya Friedman, Brian Gill, and Tadayoshi Kohno. 2010. Parenting from the pocket: Value tensions and technical directions for secure and private parent-teen mobile safety. In *Proceedings of the sixth symposium on usable privacy and security*. 1–15.
- [18] Isabella S Davis, Makayla A Thornburg, Herry Patel, and William E Pelham III. 2024. Digital Location Tracking of Children and Adolescents: A Theoretical Framework and Review. *Clinical Child and Family Psychology Review* (2024), 1–23.
- [19] Claire Elisabeth Dungey. 2024. A Bird's Eye View: Care, Control, and the Use of Surveillance Apps Among Family Members in Germany. *Sociologia* 56, 4 (2024), 325–345.
- [20] Navid Massarat Emily A. Vogels, Risa Gelles-Watnick. 2022. Teens, Social Media and Technology 2022. *Pew Research Center* (2022). <https://www.pewresearch.org/internet/2022/08/10/teens-social-media-and-technology-2022/>
- [21] Tiziano Gerosa, Lucilla Losi, and Marco Gui. 2024. The Age of the Smartphone: An Analysis of Social Predictors of Children's Age of Access and Potential Consequences Over Time. *Youth & Society* 56, 6 (2024), 1117–1143. <https://doi.org/10.1177/0044118X231223218> arXiv:<https://doi.org/10.1177/0044118X231223218>

- [22] Arup Kumar Ghosh, Charles E. Hughes, and Pamela J. Wisniewski. 2020. Circle of Trust: A New Approach to Mobile Online Safety for Families. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM, Honolulu HI USA, 1–14. <https://doi.org/10.1145/3313831.3376747>
- [23] Yasmeen Hashish, Andrea Bunt, and James E. Young. 2014. Involving children in content control: a collaborative and education-oriented content filtering approach. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '14)*. Association for Computing Machinery, New York, NY, USA, 1797–1806. <https://doi.org/10.1145/2556288.2557128>
- [24] Aaron W Kates, Huang Wu, and Chris LS Coryn. 2018. The effects of mobile phone use on academic performance: A meta-analysis. *Computers & Education* 127 (2018), 107–112.
- [25] Alex Kayal, Willem Paul Brinkman, Mark A. Neerincx, and M. Birna Van Riemsdijk. 2019. A user-centred social commitment model for location sharing applications in the family life domain. *International Journal of Agent-Oriented Software Engineering* 7, 1 (2019), 1. <https://doi.org/10.1504/IJAOSE.2019.106429>
- [26] Amanda Lenhart. 2009. Teens and Mobile Phones Over the Past Five Years: Pew Internet Looks Back. *Pew Research Center* (2009).
- [27] Mary Madden. 2017. Privacy, Security, and Digital Inequality. <https://datasociety.net/library/privacy-security-and-digital-inequality/> Publisher: Data & Society Research Institute.
- [28] Sheri Madigan, Anh Ly, Christina L. Rash, Joris Van Ouytsel, and Jeff R. Temple. 2018. Prevalence of Multiple Forms of Sexting Behavior Among Youth: A Systematic Review and Meta-analysis. *JAMA pediatrics* 172, 4 (2018), 327–335. <https://doi.org/10.1001/jamapediatrics.2017.5314>
- [29] Jörg Matthes, Marina F. Thomas, Anja Stevic, and Desirée Schmuck. 2021. Fighting over smartphones? Parents' excessive smartphone use, lack of control over children's use, and conflict. *Computers in Human Behavior* 116 (2021), 106618. <https://doi.org/10.1016/j.chb.2020.106618>
- [30] Nora McDonald, John S Seiberger, and Afsaneh Razi. 2024. For Me or Not for Me? The Ease With Which Teens Navigate Accurate and Inaccurate Personalized Social Media Content. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–7.
- [31] Kimberly J. Mitchell, Lisa M. Jones, David Finkelhor, and Janis Wolak. 2014. Trends in Unwanted Online Experiences and Sexting : Final Report. Durham, NH: Crimes against Children Research Center.
- [32] Anouk Mols, Jorge Pereira Campos, and Jason Pridmore. 2023. Family surveillance: Understanding parental monitoring, reciprocal practices, and digital resilience. *Surveillance & Society* 21, 4 (2023), 469–484.
- [33] Michael D Myers and Michael Newman. 2007. The qualitative interview in IS research: Examining the craft. *Information and organization* 17, 1 (2007), 2–26.
- [34] 1615 L. St NW, Suite 800 Washington, and DC 20036 USA 202-419-4300 | Main 202-857-8562 | Fax 202-419-4372 | Media Inquiries. 2018. A Majority of Teens Have Experienced Some Form of Cyberbullying. <https://www.pewresearch.org/internet/2018/09/27/a-majority-of-teens-have-experienced-some-form-of-cyberbullying/>
- [35] Emily Schildt, Martin Leinfors, and Louise Barkhuus. 2016. Communication, coordination and awareness around continuous location sharing. *Proceedings of the International ACM SIGGROUP Conference on Supporting Group Work* 13-16-Nove (2016), 257–265. <https://doi.org/10.1145/2957276.2957289>
- [36] Mario Schmidt. 2008. The Sankey Diagram in Energy and Material Flow Management. *Journal of Industrial Ecology* 12, 1 (2008), 82–94. <https://doi.org/10.1111/j.1530-9290.2008.00004.x> _eprint: <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1530-9290.2008.00004.x>
- [37] Marit Sukk and Andra Siibak. 2021. Caring dataveillance and the construction of “good parenting”: Estonian parents’ and pre-teens’ reflections on the use of tracking technologies. *Communications* 46, 3 (2021), 446–467. <https://doi.org/doi:10.1515/commun-2021-0045>
- [38] Pamela Wisniewski, Arup Kumar Ghosh, Heng Xu, Mary Beth Rosson, and John M. Carroll. 2017. Parental Control vs. Teen Self-Regulation: Is there a middle ground for mobile online safety?. In *Proceedings of the 2017 ACM Conference on Computer Supported Cooperative Work and Social Computing (Portland, Oregon, USA) (CSCW '17)*. Association for Computing Machinery, New York, NY, USA, 51–69. <https://doi.org/10.1145/2998181.2998352>
- [39] Pamela Wisniewski, Haiyan Jia, Heng Xu, Mary Beth Rosson, and John M. Carroll. 2015. “Preventative” vs. “Reactive”: How Parental Mediation Influences Teens’ Social Media Privacy Behaviors. In *Proceedings of the 18th ACM Conference on Computer Supported Cooperative Work & Social Computing (CSCW '15)*. Association for Computing Machinery, New York, NY, USA, 302–316. <https://doi.org/10.1145/2675133.2675293>
- [40] Xinyi Zhang, Mamtaj Akter, Heajun An, Minqian Liu, Qi Zhang, Lifu Huang, Jin-Hee Cho, Pamela J. Wisniewski, and Sang Won Lee. 2026. From Vulnerable to Resilient: Examining Parent and Teen Perceptions on How to Respond to Unwanted Cybergrooming Advances. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems*. 1–25.